Vectors and Linear Combinations

Vectors

Defn. *A matrix with one column is called a (column)* ***vector****.*

We use bold letters for vector variables, such as $\boldsymbol{x}$ and $\boldsymbol{v}$.

We sometimes write the column vector $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$ as $(3, 5)$.

Vector Operations

Vector ***addition*** is performed by adding the corresponding entries. ***Scalar multiplication*** is performed by scaling each entry. That is,

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix} \quad \textbf{and} \quad c \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} c\,u_1 \\ c\,u_2 \end{bmatrix}$$

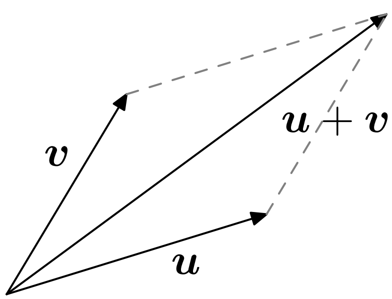
For example

$$x \begin{bmatrix} 2 \\ 4 \end{bmatrix} + y \begin{bmatrix} -1 \\ 7 \end{bmatrix} = \begin{bmatrix} 2x - y \\ 4x + 7y \end{bmatrix}$$

Vectors and Points

Defn. *We use $\mathbb{R}^d$ for the set of all d-entry vectors whose entries are real numbers.*

One can associate vector in $\mathbb{R}^d$ with the corresponding point. For example, $\mathbb{R}^2$ is the 2-dimensional plane. And vector addition can be illustrated with a parallelogram:



Linear Combinations

Defn. *A* ***linear combination*** *of vectors is formed by summing some multiple of each vector. The multipliers are called the* ***weights***.

Spans

Defn. *The* ***span*** *of a collection of vectors is the set of all possible linear combinations. If S is a set, we will denote its span by $\mathcal{S}pan\, S$.*

For example, the span of a single (nonzero) vector is a line.

The span of two vectors is (usually) a plane.

Matrix-Vector Multiplication

Defn. *If A is an $m \times n$ matrix and $\boldsymbol{x}$ is in $\mathbb{R}^n$, then the* ***matrix-vector product*** *$A\boldsymbol{x}$ is the linear combination of the columns of A specified by $\boldsymbol{x}$.*

That is, if $A = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n]$ (meaning its columns are vectors $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$), and $\boldsymbol{x} = (x_1, \ldots, x_n)$ then

$$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \ldots + x_n\boldsymbol{x}_n$$

Example of Matrix-Vector Multiplication

For example,

$$\begin{bmatrix} 2 & -1 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 4 \end{bmatrix} + 5 \begin{bmatrix} -1 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 47 \end{bmatrix}$$

Summary

A vector is a matrix with one column. We use bold letters for vector variables. $\mathbb{R}^d$ is all d-entry vectors with real entries. Vector addition adds corresponding entries; scalar multiplication scales each entry.

A linear combination of vectors is any sum of some multiple of each vector. Their span is the set of all possible linear combinations. The product of matrix A with vector $\boldsymbol{x}$ is the linear combination of columns of A given by $\boldsymbol{x}$.